# Supplementary File S2: Agent-Mediated Thermal-Twin Integration Test

This file summarizes one supervised agent-to-device test reported in the manuscript. The source record contains 21 passing assertions. The summary preserves the order and ownership of the recorded actions; it is not a complete runtime replay package.

## 1 Configuration

An isolated platform instance connected one acquisition node and three setpoint nodes to a separate gray-box extrusion/casting thermal simulator over Modbus TCP. The simulator used first-order thermal inertia and delay and was not calibrated against a physical production line. An independent Modbus client read the simulator's registers, separately from platform readback. One OMP harness with a local Qwen model operated in a manual-approval channel.

## 2 Recorded Sequence

Table 1: Distinct actions and their associated device and process evidence.

| Stage | Action and device-side record | Process and acquisition record |
|---|---|---|
| Context retrieval | The agent queried its bound nodes, the active line context, and the acquisition service. One initial query had an invalid time bound and failed; a subsequent query succeeded. | The agent retrieved the active process-value summary from the platform. |
| Agent-proposed trial | The agent submitted a three-zone recipe trial from 198 to 200 °C. Manual approval preceded dispatch. The platform receipt and independent Modbus reads confirmed all three setpoints. | The source record links the task, approval, trial receipt, and register values. The process response is not attributed to this step. |
| Follow-on approved step | A separate manually approved operation changed the three setpoints from 200 to 205 °C. Independent register reads and platform readback confirmed all three values. | Over the following 40 s, the simulated process value changed from 198.02 to 200.73 °C. Platform samples changed from 198.41 to 200.49 °C; the agent later retrieved 200.49 °C, matching the platform sample. |

## 3 Interpretation and Limits

The test demonstrates one supervised path from production-context and acquisition queries through an agent-issued recipe trial, manual approval, Modbus device writes, independent register checks, and subsequent acquisition. The reported 2.71 °C rise follows the separate 200-to-205 °C step. It is not assigned to the agent's earlier proposal. The sequential actions are not independent trials. The report does not provide a calibrated physical-process model, repeated runs, a matched controller, raw model-provider requests and responses, or a complete environment snapshot; the test therefore does not establish process-control benefit, model reasoning quality, or physical-plant performance.